

THE UNITED REPUBLIC OF TANZANIA
THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT



KIGOMA REGION
FORM FOUR MOCK EXAMINATION

(For both school and private candidates)
PHYSICS I

031

Time: 3 Hours

JULY, 2026

Instructions

1. This paper consists of sections A, B and C.
2. Answer all questions in sections A and B and any two (2) questions from section C.
3. All writings must be in **blue** or **black** ink, except for diagrams which must be in pencil.
4. Cellular phones are not allowed in the Examination room.
5. Write your Examination Number on every page of your answer booklet(s)
6. Where necessary the following constants may be used:
 - i Acceleration due to gravity $g = 10\text{N/kg}$
 - ii $\text{Pie} = 3.14$
 - iii Density of water $= 1000\text{kg/m}^3$ or (1g/cm^3)

MsomiBora.com

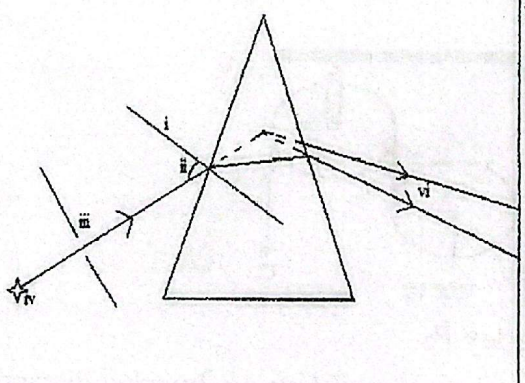
SECTION A (16 MARKS)

Answer **ALL** questions from this section

1. For each of the items (i)-(x), choose the most correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided
- i. Form three students at a certain school asked why the knowledge of physics is necessary to understand all other sciences. The research was concluded with the following points; choose the best if you were asked.
- A. Physics explains how energy passes from one object to another
 - B. Physics explains how gravity works
 - C. Physics explains motion of object that can be seen with naked eyes
 - D. Physics explains the fundamental aspects of the universe
 - E. Physics explains why object floats or sinks.
- ii. The suspended magnetic needle always comes to rest with its axis in a vertical plane called?
- A. Geographical meridian
 - B. Magnetic meridian
 - C. Geographical declination
 - D. Magnetic declination
 - E. Geographic north pole
- iii. Which of the following is the correct weight of a body of mass $24g$ when placed on the moon surface?
- A. $0.48N$
 - B. $4.8N$
 - C. $80.0N$
 - D. $0.048N$
 - E. $0.080N$
- iv. The rate of flow of charge in a conductor is
- A. Capacitance
 - B. Electric current
 - C. Potential difference
 - D. Resistance
 - E. Electromotive force
- v. Which of the following substances is not a magnetic material?
- A. Iron
 - B. Nickel
 - C. Copper
 - D. Cobalt
 - E. Steel
- vi. Why oil is used as a lubricant?
- A. It is highly viscous
 - B. It has low density
 - C. It is less viscous
 - D. It is flammable
 - E. It has high density

- vii. When charging a body by friction, which particles are transferred?
- Protons
 - Neutrons and protons
 - Neutrons
 - Electrons
 - Protons and electrons
- viii. A bus carrying a very big load on its carrier can easily overturn because
- Its centre of gravity is low
 - Its equilibrium is stable
 - Its centre of gravity is high
 - It is more stable and it can run faster
 - It cannot run faster
- ix. A light ray passing from an optically less dense medium to high dense medium
- Bends towards the normal
 - Bends away from the normal
 - Does not bend towards or away from the normal
 - Bends towards and away from the normal
 - Bends obliquely perpendicular
- x. In a concave lens parallel ray is
- Converged
 - Diverged
 - Brought to focus at the Centre of curvature of the lens
 - Transmitted parallel
 - Converged and transmitted parallel

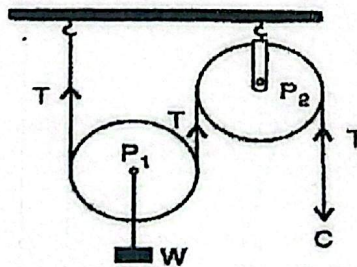
2. Match each of the descriptions of the terms used in simple machines in LIST A with the corresponding concept used in simple machines in LIST B.

List A	List B
	- Band of colours - White screen - Normal - Angle of incidence - Source of light - Slit - Angle of refractions - window

SECTION B (54 MARKS)

Answer ALL question in this section

3. (a) A convex mirror is often used as the wing mirror of a car instead of plane mirror. Justify this statement giving one point.
(b) With the aid of a diagram, show that when a plane mirror is rotated through an angle θ about an axis normal to the plane containing the incident ray and the normal, the reflected ray rotates through an angle of 2θ
4. (a) (i) Explain why the fluid at rest exerts force perpendicular to the surface?
(ii) Why the atmospheric pressure decreases rapidly not linearly?
(b) Starting from rest, the time taken by a body to slide down a 45° inclined plane with friction, is twice the time taken to slide down the same distance in the absence of friction. Determine the coefficient of friction between the body and the inclined plane.
5. (a) How would you determine the fixed points on a thermometer?
(b) Two identical free-running trolleys are on a smooth horizontal runway. One trolley is at rest and other approaches it at a constant speed of 20cm/s. after impact the two trolleys together move at 10cm/s. explain this result, stating the physical principle involved
6. (a) Explain why a feather and a coin released from the same height in an evacuated tube will reach the bottom of the tube at the same time.
(b) Rocket moves from the earth to a planet x. if it weighs 10, 000N and 30N on the earth and on planet x respectively determine the gravitation force on planet x.
7. (a) When a candle flame is brought near the cap of a charged electroscope, the electroscope discharges. Explain this observation.
(b) Two men stand a distance apart beside a long metal fence on a still day. One man places his ear against the fence, while the other gives the fences a sharp knock with a hammer two sound separated by a time interval of 0.5 seconds are heard by the first man. If the velocity of sound in air is 330m/s and in the metal is 5280m/s. How far apart are the men?
8. The diagram below shows a combination of a movable pulley P_1 with a fixed pulley P_2 used for lifting up a load W



- a) State the function of the fixed pulley P_2
- b) If the free end of the string moves through a distance x , by what distance is the load W raised?

- c) What effort has to be applied at C to just raise the load $W = 20 \text{ N}$? neglect the weight of the pulley P_1 and the friction

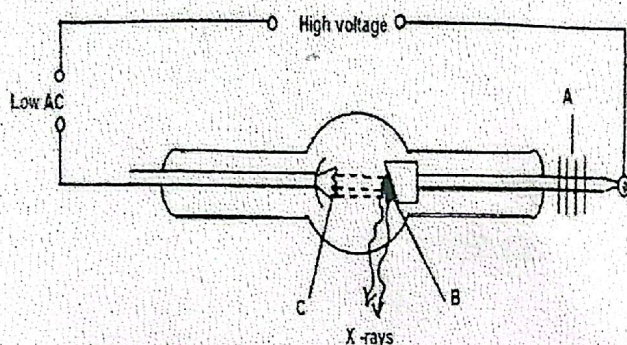
SECTION C (30 MARKS)

Answer **only** two questions

9. (a) At different times, scientists have proposed various descriptions or models of the atom to match the experimental evidence available. Briefly describe

- The plum pudding model as described by J.J. Thomson
- Planetary model as proposed by Rutherford

- (b) The figure below shows an X-rays' tube

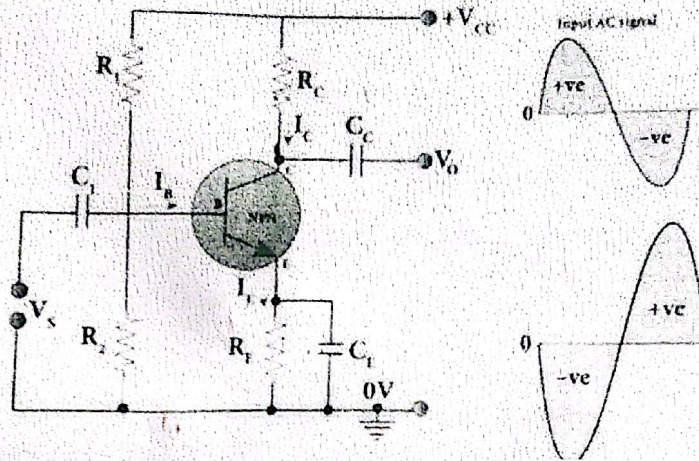


- Name the parts labeled A, B and C
 - Explain how X-Rays are produced in a tube
 - Why it is necessary to use an evacuated tube?
 - What are the purposes of high and low voltages
 - With reason, state the most appropriate metal to be used to make part B.
10. (a) Explain, using domain theory of magnetism, how an iron bar can be magnetized to saturation level by placing it in a magnetic field whose strength can be varied.
- (b) A form four student at IHOYELO secondary school was making a model of electric bell. She connected the coil in a circuit and switched it on. However, the magnetism produced in the coil was not strong enough. Then she made some changes in the coil and the circuit was now working properly. She also found out other ways of producing a strong magnetism. Describe what changes did she make in the coil? And which other ways did she discover for increasing the strength of magnetism?
- (c) i) Explain the significance of Lenz's law to show the conservation of energy in electromagnetic induction
- ii) When a speaker is brought near to a television (TV) operating using cathode ray tube, the picture on the screen is affected but when taken away, the picture becomes normal. Account for this observation.

11 (a) A galvanometer has a resistance of 5Ω and range 0-4mA. Find the resistance of the resistor which must be connected in parallel with the galvanometer if a maximum current of 10A is to be measured

b) Explain with the help of a diagram, how depletion region and potential barrier are formed in a junction diode

c). Figure below shows a NPN transistor fully stabilized



Describe the functions of:

- R_1 and R_2
- Capacitor C_E
- Capacitors C_1 and C_2
- Resistor R_E